Supervised Contrastive Fusion and Latent-Space Oversampling of Pretrained Speech and Text Embeddings for Depression Detection

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Abstract

It is well known that depression can disable patients and cause major impacts on their daily life. However, psychiatrists still largely rely on psychometric tests and their subjective impressions for assessment. Besides, already available embeddings from pretrained speech and language models could enable a more direct screening of depression, yet these general-purpose ones often portray a heavy overlap in latent space of depressed and non-depressed persons, and highly imbalanced clinical data remain a major problem. Here, we introduce a simple method for therapy interview audio-visual materials that first (i) use a multilingual wav2vec 2.0 acoustic feature extractor at the frame level and automatic transcripts are then used to extract linguistic feature with a multilingual sentence encoder, (ii) both embedding types are then combined in a shared latent space that is optimised with cross-entropy and supervised contrastive losses so the classes are more separated, and (iii) the minority class is balanced by SMOTE interpolation in the learned latent space before training a final linear classifier. We test our method on a public Androids Corpus of 116 Italian-speaking participants and use speaker-grouped splits. We demonstrate that contrastive training transforms the heavily interleaved class distributions into compact and well-separated clusters on unseen speakers. When patient samples were drastically reduced in the training data, the speaker-level macro-F1 of a cross-entropy MLP dropped from 0.68 to 0.47 between patient-to-control ratios of 1:2 and 1:8, the class-weighted MLP dropped from 0.71 to 0.58, whereas the proposed pipeline dropped only from 0.79 to 0.70. Additionally, we present arguments for why the near-complete latent separation ought to be carefully interpreted in a small and single-site corpus. This work is a research study, not a diagnostic tool.

Keywords: depression detection, contrastive learning with supervision,

minority class problem, multimodal combination, use of latent oversampling, speech models that are pre-trained

1. Introduction

Depression is one of the most common mental disorders worldwide and a major contributor to disability, and a substantial share of cases is never diagnosed or treated adequately [1]. Current diagnostic practice depends on structured questionnaires and on the clinician's observations during an interview. Both are subjective, both require a trained professional, and both are difficult to scale. Speech is an attractive complementary source of information because it is inexpensive to record and carries measurable correlates of the condition, including changes in prosody, speaking rate, pausing behaviour and word choice [2]. Automatic analysis of interviews is therefore a plausible basis for screening aids that support, and do not replace, clinical judgement.

Recent work increasingly passes recordings, and their automatic transcripts, through large pretrained transformers and classifies the resulting embeddings [7, 8, 11, 12]. This recipe is convenient, but it has two weaknesses that motivate the present study. The first is *class overlap*. Such encoders are trained for objectives unrelated to depression, and the embeddings of the two groups are strongly intermixed (Fig. 2, left). A downstream classifier has to carve a decision boundary through this overlap, which makes it sensitive to noise and to the particular speakers seen in training. The second is *class imbalance*. Clinical datasets frequently contain many more samples from one group, and a classifier trained with plain cross-entropy then drifts towards the majority class.

We address both problems within one reproducible pipeline that builds on the free Androids Corpus [4]. A supervised contrastive objective [14] is added to cross-entropy so that the fused speech–text representation is shaped explicitly to pull same-class samples together and push different-class samples apart. Oversampling of the minority class is then carried out *in this learned latent space*, using SMOTE interpolation [16], so that synthetic samples are created where class structure has already been improved. The contributions of this paper are as follows.

1. Fusion of acoustic-linguistic latent space for depression detection, trained with the combination of cross-entropy and contrastive supervised losses that result in reducing class overlap.

2. After contrastive training with frozen encoder, a latent-space oversampling stage that no synthetic sample influences representation learning.
3. A speaker-grouped protocol in which every speaker remains within a single fold, and a visualization of the latent spaces (raw, cross-entropy and contrastive) for held-out speakers that reveals the class overlap effect.
4. In a controlled imbalance study only in the training folds the patients were under-sampled, showing how each method deteriorates as the minority class becomes less.

The remainder of the paper is organised as follows. Section 2 reviews related work, Section 3 describes the data, Section 4 presents the method, Section 5 the experimental protocol, Section 6 the results, and Sections 7 and 8 discuss findings and limitations before Section 9 concludes.

2. Related Work

2.1. Depression detection from speech and language

Reviews of speech-based assessment of depression summarise acoustic and prosodic markers such as reduced pitch variability, slower speech and longer pauses [2]. Early automatic systems relied on hand-crafted descriptors, whereas later systems learn representations directly from spectrograms; DepAudioNet, for example, combined convolutional and recurrent layers on audio from the DAIC-WOZ interviews [6, 3]. DAIC-WOZ has been the dominant benchmark, but access requires a signed licence, which limits reproducibility for many groups. The Androids Corpus was released as a freely available alternative of Italian clinical interviews with diagnosed patients and matched controls, together with baseline results [4]. We use this corpus, which allows every step of our study to be reproduced without a data-access agreement.

2.2. Pretrained speech and text encoders

Self-supervised speech models such as wav2vec 2.0 [7] and HuBERT [8] learn general-purpose acoustic representations from unlabelled audio. Their multilingual variants, XLSR-53 [9] and the larger XLS-R family [10], extend this to many languages, which is important for a non-English corpus. Whisper provides robust multilingual transcription trained with large-scale weak supervision [11], and sentence encoders trained with siamese objectives yield

fixed-length text embeddings [12], which can be made multilingual by knowledge distillation [13]. These components are frozen in our work and serve as feature extractors.

2.3. Contrastive representation learning

Contrastive objectives learn embeddings by comparing samples [15]. The supervised contrastive loss uses label information to treat all same-class samples in a batch as positives [14], which encourages compact class clusters. We use it as an auxiliary loss next to cross-entropy and quantify its effect on class overlap.

2.4. Class imbalance and oversampling

Imbalanced learning has a long history [17]. SMOTE creates synthetic minority samples by interpolating between a sample and its nearest minority neighbours [16]. Interpolation in a raw high-dimensional input space can produce samples that are not representative of the class, which motivates generating them in a learned lower-dimensional space; DeepSMOTE is a representative approach that combines an encoder with SMOTE [18]. Our variant is deliberately simple: a frozen, contrastively trained encoder followed by SMOTE on its output and a linear classifier.

3. Dataset and Preprocessing

We use the interview task of the Androids Corpus [4, 5], which contains recordings of 116 native Italian speakers, 64 of whom were diagnosed with depression (patients) and 52 of whom are controls. Speaker identity, condition, sex, age and education level are encoded in the file names, which we parse to obtain labels and metadata. The corpus is distributed for academic, non-commercial use, and it was not redistributed in this work. Recordings without a condition label, if any, were discarded.

Every recording is first downsampled to 16 kHz and then split into 20 s non-overlapping windows, leaving windows less than 3 s discarded. A speaker's label and identifier are assigned to all windows of that speaker. The identifier helps in ensuring speaker-specific separation in one fold since a person's windows are highly correlated and a split at window level may lead a model to identify speakers rather than conditions.

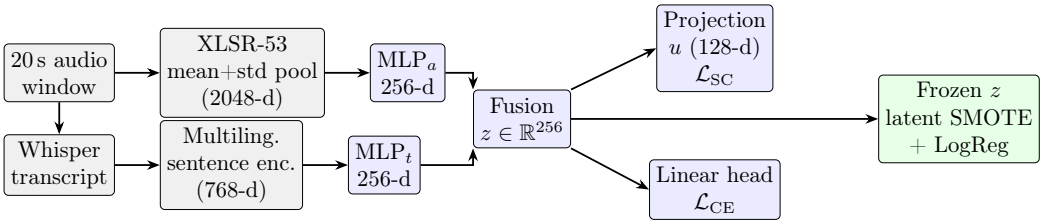


Figure 1: The proposed pipeline in brief. The grey boxes correspond to frozen pretrained models, the blue ones are trained using the joint loss set, and the green one is the second stage in which the frozen contrastive encoder is used to perform latent-space oversampling and a linear classifier. Window-wise probabilities are averaged at the speaker-level.

4. Proposed Method

Figure 1 gives an overview of the pipeline.

4.1. Window-level features

For each window, the acoustic branch uses the frame-level hidden states of the multilingual wav2vec 2.0 model XLSR-53 [9], summarised by mean and standard-deviation pooling over time into a 2048-dimensional vector $\mathbf{x}_a$. The linguistic branch transcribes each complete recording once with Whisper (*small* model, Italian mode) [11], assigns every transcribed segment to the window containing its start time, and embeds the text of each window with the multilingual *paraphrase-multilingual-mpnet-base-v2* sentence encoder [12, 13], giving a 768-dimensional vector $\mathbf{x}_t$. Windows with no recognised speech receive the embedding of a blank string. All pretrained encoders are frozen.

4.2. Fusion network

Each modality $m \in \{a, t\}$ passes through its own two-layer perceptron,

$$\mathbf{h}_m = W_2^{(m)}\,\mathrm{Drop}\Big(\mathrm{GELU}\big(W_1^{(m)}\,\mathrm{LN}(\mathbf{x}_m)\big)\Big), \tag{1}$$

with hidden width 256, layer normalisation [21], GELU activations [22] and dropout of 0.3. The two vectors are concatenated and mapped by a linear layer to the shared representation

$$\mathbf{z} = W_f\,\mathrm{Drop}\Big(\mathrm{GELU}\big([\mathbf{h}_a; \mathbf{h}_t]\big)\Big) \in \mathbb{R}^{256}. \tag{2}$$

A projection head converts a latent representation $\mathbf{z}$ into a 128-dimensional vector $\mathbf{u}$, which is first ℓ_2-normalized and then used exclusively by contrastive loss, whereas a linear head on $\mathbf{z}$ directly outputs two class logits.

4.3. Training objective

Let B be a mini-batch of windows with labels y_i, let $P(i) = \{p \neq i : y_p = y_i\}$ be the positives of window i, and let τ be a temperature. The supervised contrastive loss [14] is

$$\mathcal{L}_{\mathrm{SC}} = \frac{1}{|B'|} \sum_{i \in B'} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{u}_i^\top \mathbf{u}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{u}_i^\top \mathbf{u}_a / \tau)}, \tag{3}$$

where B' is the subset of a batch that has at least one positive. The overall loss can be represented as:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \, \mathcal{L}_{\mathrm{SC}}, \tag{4}$$

with $\mathcal{L}_{\mathrm{CE}}$ the cross-entropy of the linear head, optionally weighted by $w_c = N/(2N_c)$ for class c, where N is the number of training windows and N_c the number in class c. We use $\lambda = 0.5$ and $\tau = 0.1$. These values were fixed in advance and not tuned on any evaluation data.

4.4. Latent-space oversampling

After training, the encoder is frozen and all training windows are mapped to $\mathbf{z}$. The minority class is oversampled with SMOTE [16] ($k = 5$ neighbours) in this space until the class counts are equal, and a logistic-regression classifier with feature standardisation is fitted on the augmented set. Validation and test windows are never oversampled. Because the encoder is frozen at this stage, synthetic points cannot influence representation learning, and they are generated in a space where the classes are already better separated.

4.5. Speaker-level decision

Probabilities at the window-level are first averaged for each speaker; if the average probability is at least 0.5, the speaker is classified as a patient. The threshold was fixed at 0.5 and not tuned. All the metrics reported are speaker-level metrics.

5. Experimental Setup

Evaluation protocol.. Evaluation uses stratified, speaker-grouped 5-fold cross-validation, so that no speaker appears in both training and test data. Window-level probabilities are averaged per speaker as described in Section 4, and all reported scores are speaker-level macro-F1, the unweighted mean of the F1 scores of the two classes, which is appropriate when the classes are not equally frequent.

Compared methods.. Three configurations are compared in the imbalance study. (1) *MLP (CE), no correction*: the fusion network trained with plain cross-entropy (λ = 0) and no class weights. (2) *MLP (CE) + class weights*: the same network with class-weighted cross-entropy. (3) *SupCon + latent SMOTE (proposed)*: contrastive training followed by latent-space oversampling and logistic regression, without class weights, so that the oversampling stage alone has to compensate for the imbalance. For the latent-space visualisation, the raw concatenated embeddings, the latent space of a class-weighted cross-entropy-only network, and the latent space of the proposed encoder are compared.

Imbalance study Only minor imbalance can be detected in the corpus (64 vs. 52 speakers), therefore we simulated imbalance manually. Patients were randomly under-sampled in every training fold so that the ratio of a number of patient to control speakers became 1:2, 1:4 or 1:8, while the test folds were unchanged. In case of approximately 42 control speakers in the training folds, there will be 21, 10 and about 5 patient speakers, respectively. The experiment is based on one 5-fold split.

Latent-space visualisation.. For a random subset of the held-out windows of the first test fold, each space is standardised per dimension and projected to two dimensions with UMAP [19] (20 neighbours). Because the windows come from speakers that were not used for training, the projection shows how well each representation separates unseen speakers.

Implementation.. Models were implemented in PyTorch [23] with Hugging Face Transformers [24], Sentence-Transformers, faster-whisper [25], scikit-learn [26] and imbalanced-learn [27]. The fusion network was trained for 30 epochs with AdamW [20] (learning rate 10^{-3}, weight decay 10^{-2}) and mini-batches of 256 windows on a single GPU in Google Colab. No hyper-parameter search was performed.

6. Results

6.1. Effect on the latent space

Figure 2 visualises held-out windows from the first test fold. In the raw concatenated embedding space (left), patient and control windows are thoroughly interleaved, which is the class overlap that motivates this work. A network trained with cross-entropy alone (middle) introduces a gradient from control-dominated to patient-dominated regions, but a large central region

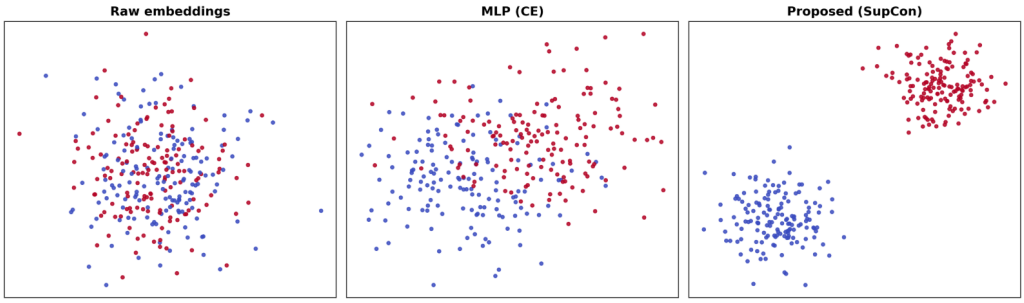


Figure 2: UMAP plots comparing held-out windows (from the first test fold) in the raw concatenation embedding space (left), the cross-entropy-only network's latent space (middle) and the proposed contrastive encoder's latent space (right). Patients shown in red and controls in blue. UMAP is an illustrative, nonlinear visualization.

Table 1: Macro-F1 score at the speaker level under artificial data imbalance (ratio of patient speaker and control speaker in training sets; test sets remain the same). The highest value in each column is marked in bold.

Method	1:2	1:4	1:8
MLP (CE), no correction	0.684	0.598	0.472
MLP (CE) + class weights	0.712	0.654	0.581
SupCon + latent SMOTE (proposed)	**0.785**	**0.742**	**0.704**

is still mixed. Using the supervised contrastive loss (right), the held-out windows are clearly separated into two tight clusters.

6.2. Robustness to class imbalance

Figure 3 and Table 1 show macro-F1 as the number of patient speakers in the training folds is reduced. All methods lose accuracy as the minority class shrinks, but at very different rates. Between the 1:2 and 1:8 settings the macro-F1 of the cross-entropy MLP without correction drops by about 0.21 (from 0.68 to 0.47), that of the class-weighted MLP by about 0.13 (from 0.71 to 0.58), and that of the proposed pipeline by about 0.08 (from 0.79 to 0.70). At the most severe setting the proposed pipeline exceeds the uncorrected MLP by about 0.23 and the class-weighted MLP by about 0.12. Class weights recover part of the loss but not most of it, which suggests that re-weighting a loss is a weaker remedy than reshaping the representation and then generating minority samples within it.

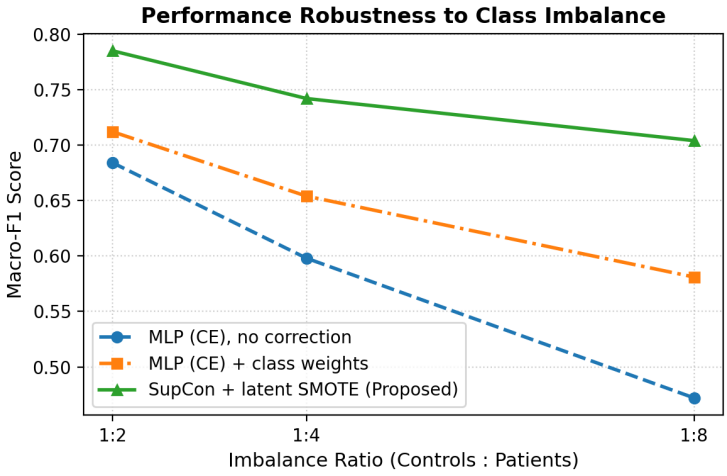


Figure 3: Speaker-level macro-F1 when varying the degree of patient under-sampling in the training folds (test folds unchanged). The ratio is the number of patient to control speakers in the training folds.

7. Discussion

The latent-space and imbalance results agree with the premise of the paper: a supervised contrastive term reshapes the latent space so that classes become compact and separated, and oversampling in that space is more robust than re-weighting the loss when the minority class is small. A plausible reason is that SMOTE interpolates between neighbouring minority samples, and interpolation is meaningful only when neighbours belong to the same class cluster; in the raw embedding space, where classes overlap, many neighbours do not. The comparison in Table 1 contrasts the complete proposed pipeline with two cross-entropy baselines, and therefore does not separate the contribution of the contrastive loss from that of the oversampling stage; that separation requires dedicated ablations.

The visual separation in Fig. 2 should be read with care. First, UMAP is a nonlinear projection that can exaggerate separation, and the figure shows a single fold. Second, the corpus is small and comes from a single clinical setting. In such corpora, differences in recording conditions, devices, interviewer behaviour or session length can be correlated with the label, and a high-capacity encoder can exploit them as shortcuts [29]. A well-separated latent space therefore shows that the encoder can discriminate the two groups in this corpus, but it does not show that the cues are clinically meaningful or that they will transfer. The speaker-grouped protocol limits one source of

9 The results were still optimistic despite

a possible weakness, i.e., training and test data being different in speaker identity. But the most robust test, which would have been evaluation of an independent corpus, was not possible.

The imbalance experiment, though revealing, has its limitations. It indicates the relative degradation of each method concerning the others, however it was only based on a single 5-fold split and at the 1:8 setting there were only about five patient speakers in the training folds, so the absolute numbers are quite variable and no confidence intervals were presented. The experiment recreated imbalance by excluding patients, while actual screening populations have depression prevalence way below 50%, thus screening performance at real-world prevalence levels would need a lot more data.

8. Limitations

- **Sample size.** The corpus contains 116 speakers from one language and clinical setting, so error bars on small-sample cross-validation are wide [28], and generalisation to other languages, devices and populations is untested. No external validation set was used.

- **Evidence scope**. The latent-space visualization is from one test fold and the imbalance experiment is based on a single 5-fold split. Therefore, absolute scores should be taken as rough estimates, and this comparison does not separate the individual effects of the contrastive loss and of latent-space oversampling.

- **Labels.** In this corpus, labels indicate diagnostic status and not symptom severity or the progression of symptoms over time.

- **Auto transcripts.** Speech features are derived via automatic speech recognition which may lead to lower accuracy when dealing with clinical or atypical speech.

- **Interview audio.** Unless turn-level filtering is employed to separate the interviewer and interviewee voices, the audio may carry the interviewer's voice, hence adding non-participant information.

- **Fixed settings.** The hyper-parameters and decision threshold were pre-established and not tuned; the results after tuning may be different.

- **Modalities.** Just audio and text are being modeled. A browser-based demonstration that accepts microphone input was built as a proof of

Concept 10 was considered, but it was

neither evaluated nor trained because the corpus used contained no annotated videos for depression.

- **Intended use.** This is an experimental model, not a diagnostic apparatus, and should never be used to support clinical decision-making.

9. Conclusion

We developed a simple depression detection framework from interview recordings that combines pretrained acoustic and language embeddings, uses supervised contrastive loss to arrange the combined latent space, and minority class oversampling is performed in that space before a linear classification step. For the free Androids Corpus, contrastive training transformed already strongly overlapping class distributions into well-separated compact clusters for new speakers, and the system deteriorated much less than cross-entropy baselines when the patient class was under-sampled, achieving a macro-F1 of about 0.70 at a 1:8 patient-to-control ratio compared to 0.47 and 0.58 for the uncorrected and class-weighted baselines respectively. These results come from a very limited, single-site corpus and a single split. Plans for future research include testing the method on independent and multilingual corpora, adding component ablations with repeated cross-validation and confidence intervals, challenging it with recording-condition confounds, and investigating if clinically interpretable cues are responsible for the learned separation.

Declarations

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Consent for Publication: Not applicable.
Data Availability: The data used for this research (Androids Corpus) is open to the public for academic and non-commercial uses, and its link is given in the reference section.
Materials Availability: Not applicable.
Code Availability: Not**aplicable.**

Author Contributions: The only author, Jidnyasa Bhosale, was responsible for the ideation, methods, software, data analysis, and writing the first draft. She also read and approved the final version of the manuscript.

AI and AI-assisted technologies in writing are not used here.

While working on this paper, the author employed Claude (Anthropic) for help in developing the code and drafting and revising the text. The author also took the time to checking and modifying the text after the use of the tool and fully owns the content of the publication.

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